Assignment #4: Microbiome diversity in Colon Cancer Environment

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BINF*6210

December 5, 2025

Git Hub: https://github.com/hannahxg20/BINF6210_Assignment_4/tree/main

e taken from https://burjeel.com/colon-cancer-treatment/

Colorectal Cancer (CR)

- Second leading cause of cancer-related deaths, and mortality from CRC has been rising since 2000[1,2]
- Development influenced by genetic, environmental, and lifestyle factors [1,3]

Gut Microbiome & CRC

- Gut microbiome support intestinal homeostasis through cell renewal, gut barrier maintenance, and immune regulation[4,5]
- Individuals with CRC (microbial dysbiosis) exhibit altered gut microbiome relative to healthy populations[6]

= Biodiversity and composition of your gut microbes matters!

For example, meta-analyses demonstrated poorer survival in CRC patients with high levels of *Fusobacterium nucleatum* present[7]

= Gut microbiota are associated with CRC survival[8]

Studying the CRC microbiome could reveal markers for diagnosis or prognosis and uncover new treatment targets[7,9]

Research Rationale

Microbiome profiling studies in CRC have produced inconsistent results, likely driven by variations in study design and analytical methods[10] ...

- Stool vs tumor[11,12]
- Focused entirely on tumor tissue[13]
- Treated tumor and adjacent normal tissue as identical[14]

Current research focus: Does local or global regulation of microbiome drives tumorigenesis?

Exploratory Project

#Q1:**Within the colorectal cancer (CRC) environment, do tumor tissues show different biodiversity compared to adjacent normal tissues?**

#Q2 (Secondary): **Are these patterns consistent across studies?**

Data Set for this Project

Sequence Read Archive (SRA) on NCBI website

Search input “(colorectal cancer AND homo sapiens AND 16S)”

Search Criteria

- Had to be published = credible
- Between 2020 - now
- Paired-end sequencing (e.g. Illumina)
- 16S rRNA sequencing (V3-V4 region)
- Human biopsy sample
- Study with 30+ samples

2 studies met criteria (see next slide)

Downloading Data Set

- Data downloaded on Dec 1, 2025
- Used Sequence Read Archive (SRA) Toolkit to downloads the FASTQ DNA sequence files (zip files located in “FASTAQ_Data” folder)

Randomly took 10 samples

- 5 from tumor biopsies
- 5 from normal adjecent tissues

=20 samples total (see next slide)

*FASTQ files are too big to upload to Dropbox, Courselink. Files are available on GitHub repository: https://github.com/hannahxg20/BINF6210_Assignment_4/tree/main

Studies & Samples

Article | Open access | Published: 04 April 2025

Microbiome and biofilm insights from normal vs tumor tissues in Thai colorectal cancer patients

Pirada Yincharoen, Auemphon Mordmuang, Tachpon Techarang, Panus Tangngamsakul, Panchaphon Kaewubon, Paijit Atipairin, Sorawat Janwanitchasthaporn, Lavanya Goodla & Kulwadee Karnjana

npj Precision Oncology 9, Article number: 98 (2025) | Cite this article

Study 1 - SRP566493

Gut Microbes. 2024 Jun 11;16(1):2363012. doi: 10.1080/19490976.2024.2363012

Association between gut microbiota and CpG island methylator phenotype in colorectal cancer

Study 2 -SRP449898

Pyoung Hwa Park [a,b], Kelsey Keith [a,b], Gennaro Calendo [b], Jaroslav Jelinek [a,b,c], Jozef Madzo [a,b,c], Raad Z Gharaibeh [d,e], Jayashri Ghosh [a], Carmen Sapienza [a], Christian Jobin [d], Jean-Pierre J Issa [a,b,c]

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PMCID: PMC11174071 PMID: 38860458

*Please see "SraAccessionList" Excel spreadsheet.

Study	Individual Sample	Biopsy Type	Platform	Sequence
SRP566493	SRR32497531	Normal	Illumina	Paired
SRP566493	SRR32497532	Normal	Illumina	Paired
SRP566493	SRR32497533	Normal	Illumina	Paired
SRP566493	SRR32497534	Normal	Illumina	Paired
SRP566493	SRR32497535	Normal	Illumina	Paired
SRP566493	SRR32497540	CRC	Illumina	Paired
SRP566493	SRR32497541	CRC	Illumina	Paired
SRP566493	SRR32497542	CRC	Illumina	Paired
SRP566493	SRR32497543	CRC	Illumina	Paired
SRP566493	SRR32497544	CRC	Illumina	Paired
SRP449898	SRR25305917	Normal	Illumina	Paired
SRP449898	SRR25305918	Normal	Illumina	Paired
SRP449898	SRR25305919	Normal	Illumina	Paired
SRP449898	SRR25305920	Normal	Illumina	Paired
SRP449898	SRR25305921	Normal	Illumina	Paired
SRP449898	SRR25374795	CRC	Illumina	Paired
SRP449898	SRR25305896	CRC	Illumina	Paired
SRP449898	SRR25305897	CRC	Illumina	Paired
SRP449898	SRR25305898	CRC	Illumina	Paired
SRP449898	SRR25305899	CRC	Illumina	Paired

Pipeline: Methods

Saved in "RDS_objects" folder

Quality Control:

- ✓ seqtab_nochim.rds
- ✓ track.rds
- ✓ sample_names.rds
- ✓ asv_sequences.rds
- ✓ filtering_stats.rds

Assigning Taxonomy:

- ✓ taxa.rds

Create Phylogenetic Tree:

- ✓ tree.rds

Metadata & Phyloseq:

- ✓ metadata.rds
- ✓ ps.rds

Alpha Diversity metrics:

- ✓ diversity_metrics_results.rds

Quality Control using DADA2

- Trim & Filter reads
- Error Rate Estimation
- Merge reads
- Chimera removal

Assign Taxonomy with Decipher

- Used SILVA_SSU_r138_2_2024.Rdata
- Assigned and extracted taxonomy

Create Phylogenetic Tree with Decipher & phangorn

- Neighbor joining only
- Extract ASV sequences
- Alignment

Metadata & Phyloseq

- Created metadata to match sample names
- Built Phyloseq

Calculate Alpha Diversity Metrics

- Create and align OUT matrix with phylogenetic tree
- Calculate species richness, Shannon Diversity, Faith's Phylogenetic Diversity (PD)
- Verify all vectors have same length & combined all results

Statistical tests

- Wilcoxon Test: Richness, Shannon Diversity, Faith's PD
- Compare between Study 1 & Study 2 (tumor vs normal tissue)
- Mean difference in richness (Study 1, Study 2)
- Permutation test for adonis under reduced model

Create Visualization

- Richness by Tissue Type
- Shannon Diversity by Tissue Type
- Faith's PD by Tissue Type
- Richness by Study
- NMDS Ordination

Script run required ~4 hours, so I created an **RDS_objects** folder to store all necessary files and added "save checkpoints" after each step. Each step now also loads required objects if they're not already in memory.

Figure 1

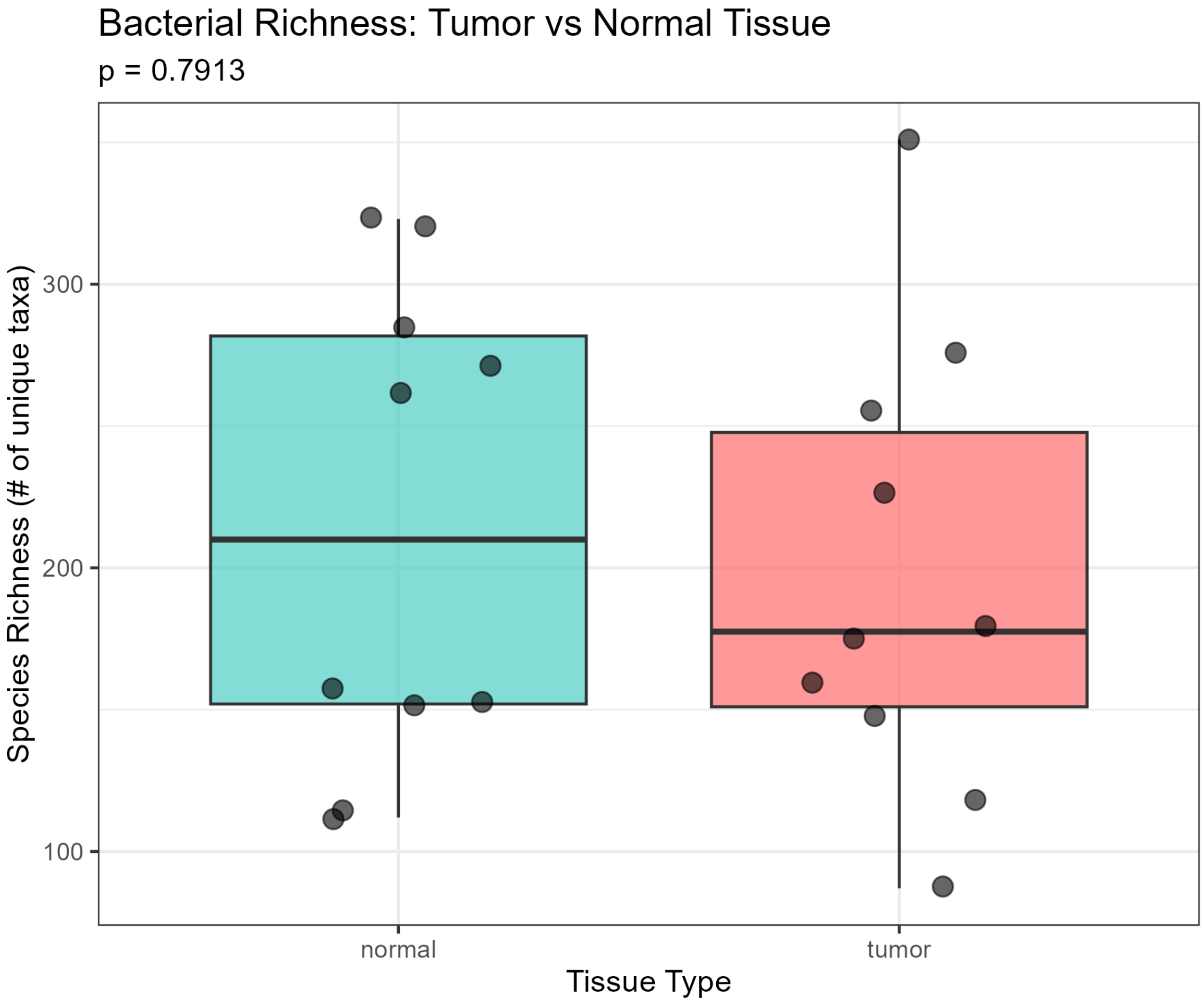

Bacterial Richness: Tumor vs Normal Tissue
p = 0.7913
Species Richness (# of unique taxa)
300
200
100
normal
tumor
Tissue Type

Figure 2

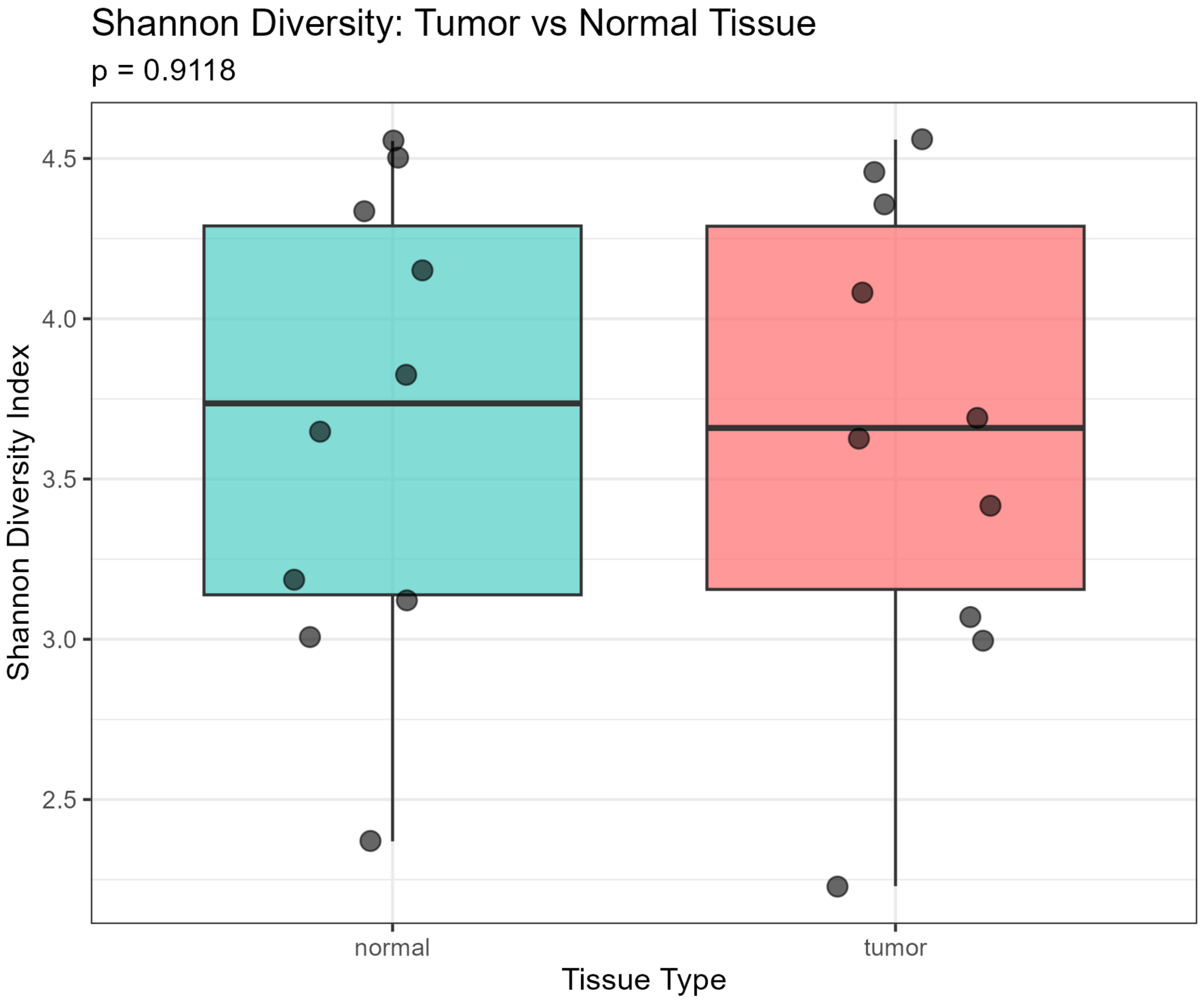
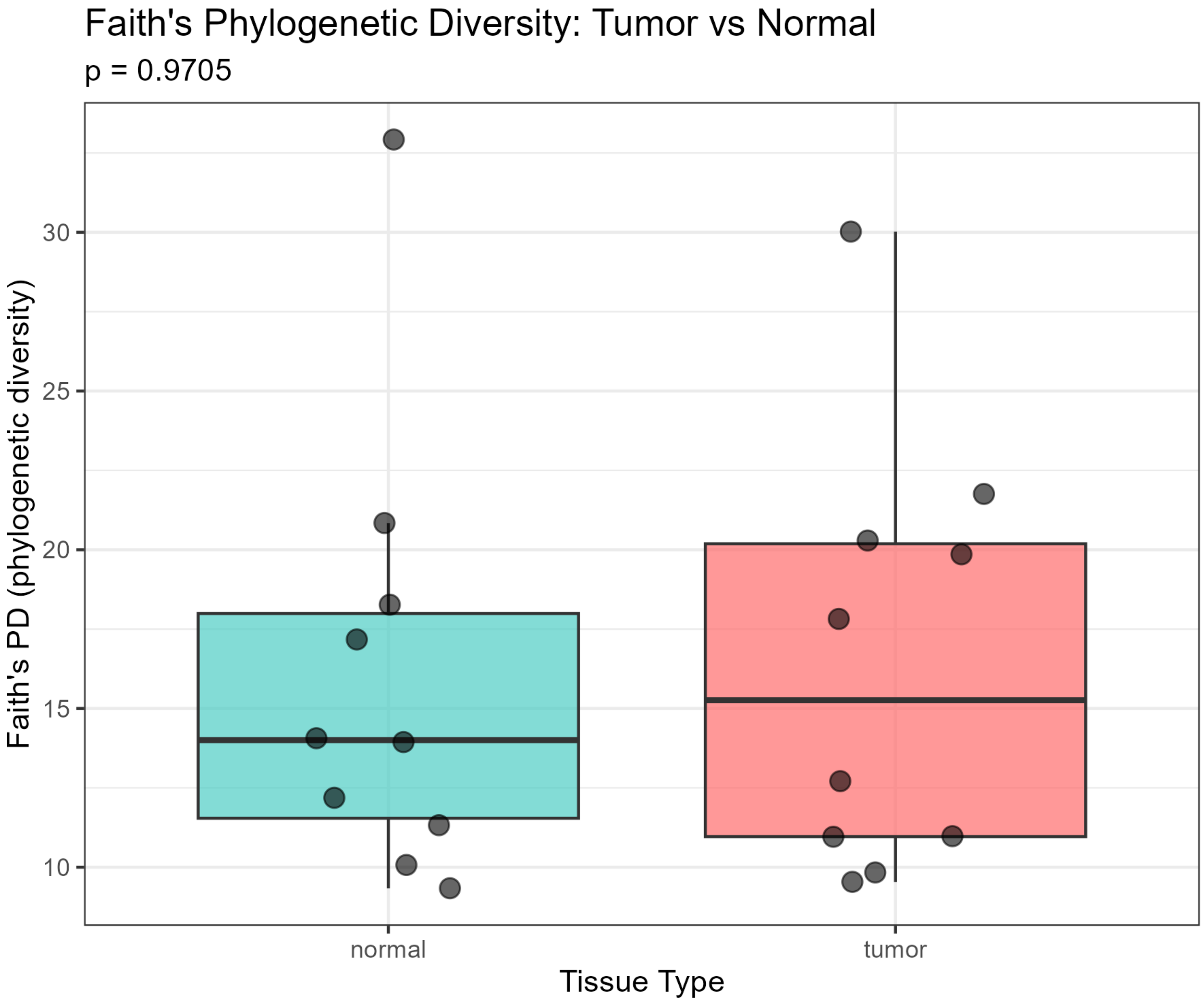

Shannon Diversity: Tumor vs Normal Tissue
p = 0.9118
Shannon Diversity Index
4.5
4.0
3.5
3.0
2.5
normal
tumor
Tissue Type

Figure 3



Faith's Phylogenetic Diversity: Tumor vs Normal
p = 0.9705
Faith's PD (phylogenetic diversity)
30
25
20
15
10
normal
tumor
Tissue Type

Figure 4

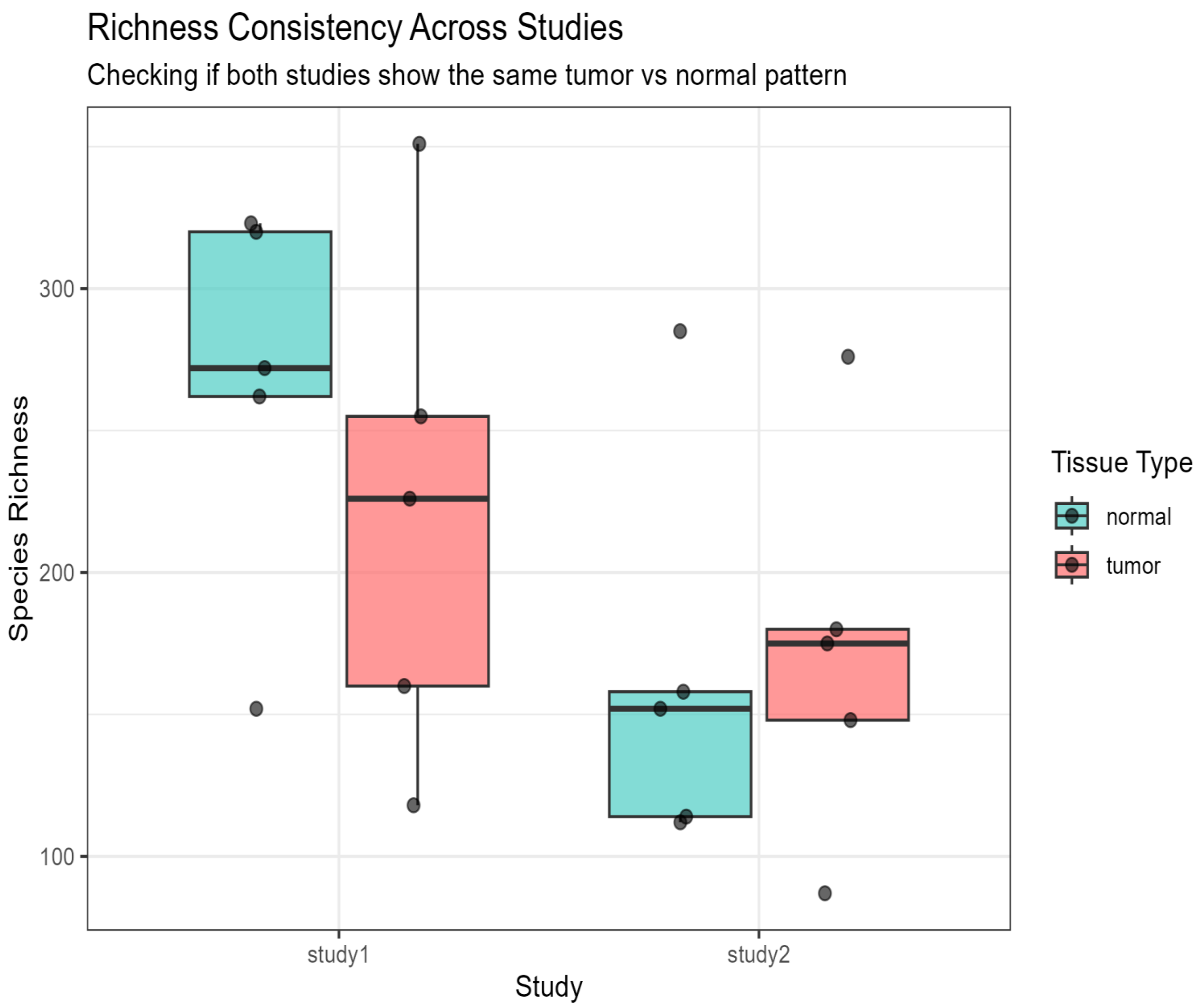
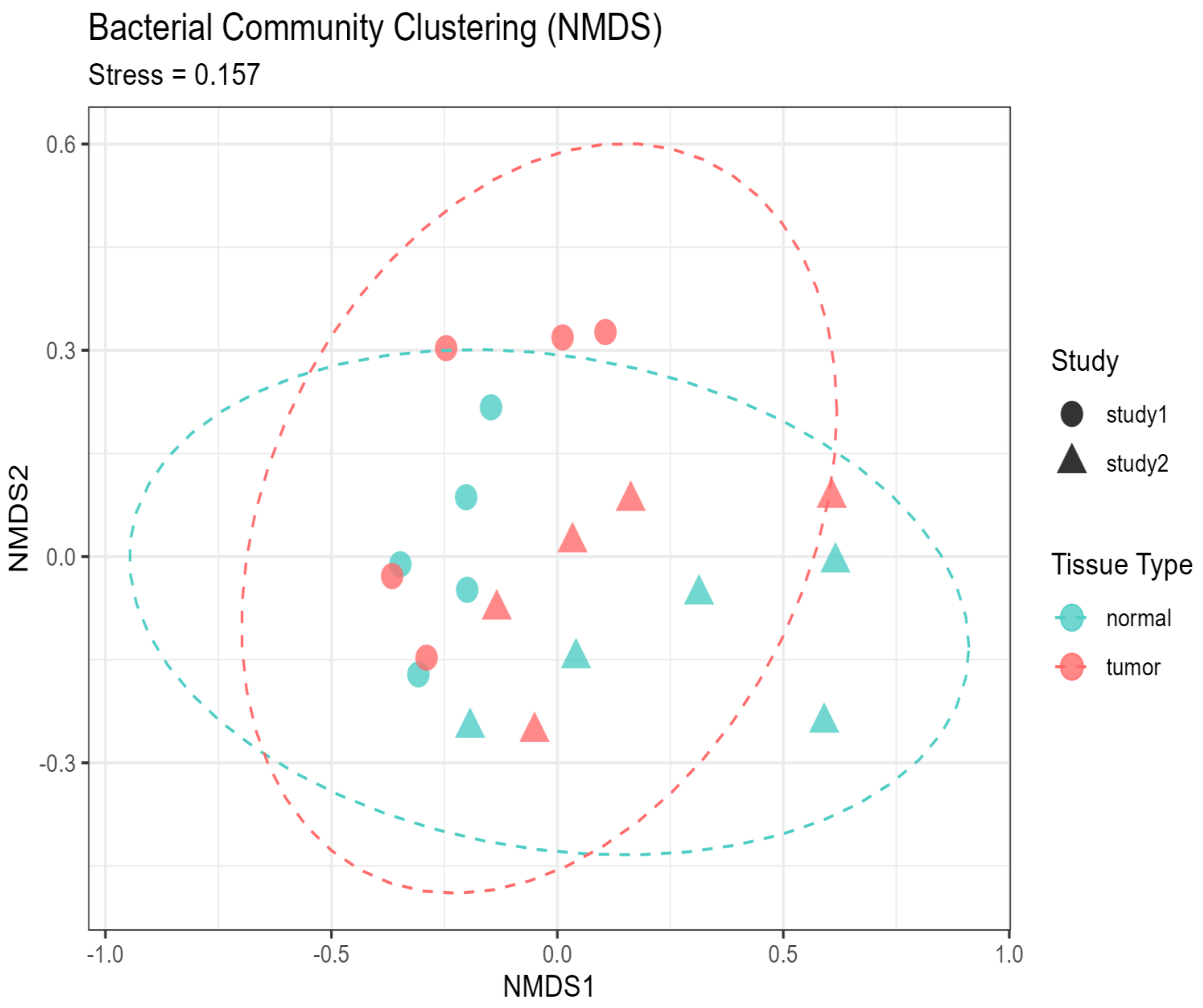

Richness Consistency Across Studies
Checking if both studies show the same tumor vs normal pattern
Species Richness
300
200
100
study1
study2
Study
Tissue Type
normal
tumor

Figure 5



Bacterial Community Clustering (NMDS)
Stress = 0.157
NMDS2
0.6
0.3
0.0
-0.3
-1.0
-0.5
0.0
0.5
1.0
NMDS1
Study
study1
study2
Tissue Type
normal
tumor

Results & Discussion

Bacterial diversity (*Figure 1-3*)

No significant difference between tumor and adjacent normal CRC tissues (Wilcoxon test, all $p > 0.05$)

- Richness: $p = 0.79$
- Shannon diversity: $p = 0.91$
- Faith's phylogenetic diversity (PD): $p=0.97$

= suggests that the tumor microenvironment does not substantially alter the overall diversity of the bacterial community in these samples

Results & Discussion

Mean value richness (*Figure 4*)

Patterns were inconsistent across studies, with Study 1 showing lower richness in tumors and Study 2 showing higher richness in tumors, neither reaching statistical significance.

- Study 1 (SRP566493): Tumor = 222 vs Normal = 265.8 (tumor has LOWER richness)
- Study 2 (SRP449898): Tumor = 173.2 vs Normal = 164.2 (tumor has HIGHER richness)

NMDS Plot (*Figure 5*) & PERMANOVA results

Plot accurately represents the differences (Stress = 0.157)

PERMANOVA ($p = 0.849$)

= Tumor and normal CRC tissues have similar bacterial communities.

Summary & Comparison to Literature

Summary

Q1: Within the colorectal cancer (CRC) environment, do tumor tissues show different biodiversity compared to adjacent normal tissues?

NO - Across all tests:

- ✗ Richness: p = 0.79 (not different)
- ✗ Shannon: p = 0.91 (not different)
- ✗ Faith's PD: p = 0.97 (not different)
- ✗ Community composition (PERMANOVA): p = 0.85 (not different)

Q2: Are patterns consistent across studies?

Patterns are inconsistent:

- Study 1: Tumor has lower richness than normal
- Study 2: Tumor has higher richness than normal
- Neither study shows significant differences

What is seen in the literature?

- Microbiomes of tumor and normal tissue in CRC individuals differ [15,16]
- Appears to be the consensus in the field[15–19]

Project's Limitations & Improvements

Possible reasons WHY there is difference in results between this project and current literature

- Sample size is too small to detect any difference
- Unable to pair biopsies from the same patient in order to maintain patient confidentiality.
- Study-specific factors might influence results (e.g. study did not record tumor location – known to have some differences in microbiota) [17,19]

How to improve this project?

- Increase to 30+ samples per group (normal vs tumor tissue)
- Use paired samples from same patient (normal and tumor tissue biopsy)
- Include samples from individuals that do not have CRC
- Deeper community analysis and statistical analysis (e.g. linear mixed models, differential abundance testing)
- Collect other information besides tissue type and study ID (e.g. patient demographics, diet information)

Reflection

Through this small project I learned how to download FASTQ files from Sequence Read Archive (SRA) on the NCBI website, I also learned how to navigate through the DADA2 pipeline and how to assign taxonomy with decipher. I also developed basic skills on how to create a phylogenetic tree and perform diversity calculations on R studio.

In this course, I learned how to apply critical thinking from a bioinformatician standpoint and gained several tips on coding best practices, productivity tools (e.g. GitHub), and how to obtain information from databases (e.g. BOLD).

I hope to apply this knowledge (microbiome data analysis on R studio) to my career goals of intestinal health research. This is definitely one possible direction my current laboratory research could go!

Acknowledgements

I would like to thank our TA, Saira Asif, for help on downloading FASTQ files from SRA.

I would like to thank Benjamin Callahan (author of the DADA2 Pipeline Tutorial (1.16)), NCBI GitHub team, and the NIH Bioinformatics Training and Education Program for creating their online tutorials.

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